FacilityCore — data model and flows

Product Phase 1. One schema, drawn in parts so a page stays readable. The overview is the whole model. The six diagrams after it carry the columns. The sequences are the flows to build.

Print this to PDF from a preview that draws Mermaid. A plain text converter will show the diagram source instead of the picture. Landscape for the overview.

Money is integer paise. A positive ledger amount means the flat owes the society. Dates and times are stored in UTC and shown in Asia/Kolkata.

Rules the schema enforces

- One database, many societies. Every business row carries `society_id`. A query that can see a second society is a bug.
- `APP_USER` is global, keyed by mobile. A system admin has a role row with no society. Everyone else acts inside one society per session.
- A session token carries one role. That token cannot call another role's actions.
- Guards have no account. SOS uses `SOCIETY.duty_guard_mobile`.
- One flat has one open bill payer. Extra phones on the flat can view, pay, and complain. Only the bill payer is reminded.
- One flat has at most one bill for a given month. Overdue is calculated from the balance and the due date. It is not a stored balance and not a bill status.
- A sent bill is never edited in place. A correction is a new adjustment line plus a ledger row, with the person, the time, and the reason.
- Payment rows are never deleted. A mistake is a reversing ledger row and a new row.
- A UPI reference on a reported or confirmed payment cannot be used again. A rejected reference can be sent again.
- Late fees pause only while a payment is `reported`. A rejection accrues the waiting days. Confirm stops the fee.
- Incoming money clears late fees first, then the oldest bill. Extra money is an advance (negative balance) or a recorded refund.
- An interior-work deposit has its own rows. It never posts to `LEDGER_ENTRY`.
- `PROVIDER_PAYMENT` has no foreign key to `BILL`, `LEDGER_ENTRY`, or `PAYMENT`. Service money cannot reduce dues. There is no screen for it in this phase.
- `AUDIT_EVENT` and `LEDGER_ENTRY` are append-only.
- There is no vendor, phonebook, visitor, face-template, gateway, or commission table.

Role names below use Secretary for the BRD's Builder/Admin role.

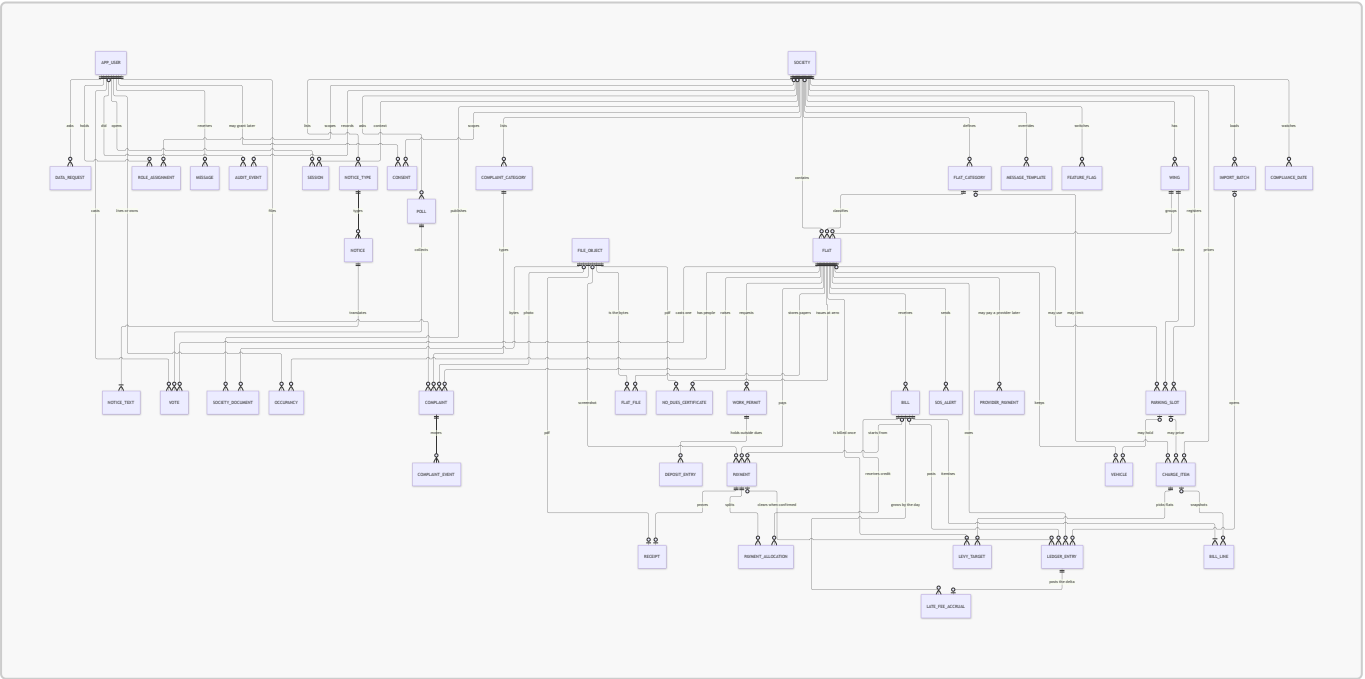
Action	System admin	Secretary	Treasurer	Facility manager	Resident
Create a society, set Draft / Active / Suspended, grant or remove a committee role	yes				

Action	System admin	Secretary	Treasurer	Facility manager	Resident
Flats, import, opening balances, charges, settings, handover, notices, polls, documents		yes			
Send bills, record cash or cheque, defaulter list, reports, money history		yes	yes		
Confirm or reject a reported UPI payment			yes		
Waive a late fee, and only with a reason		yes			
Complaints		yes		yes	own flat
Vote, read notices and society documents, SOS, own statement					yes
Apply a provider payment to dues	no	no	no	no	no

While a society is **Draft**, the secretary can sign in and load data. Residents cannot. **Suspended** revokes every session for that society. System admin is outside the society and can still open it.

1. Overview

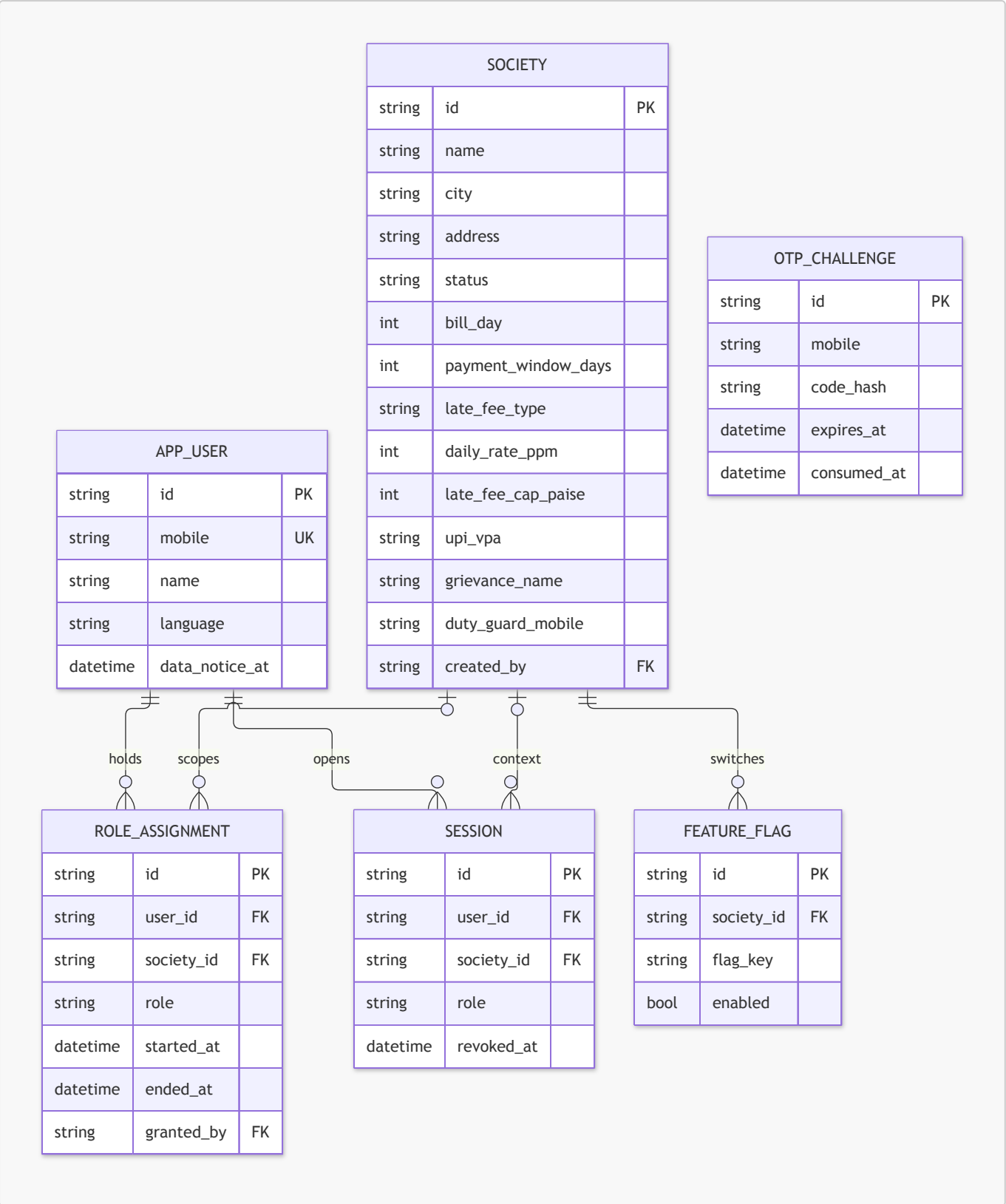
No columns here. The later diagrams define them. `PROVIDER_PAYMENT` is on this map and has no line to `BILL` or `LEDGER_ENTRY`.



2. Identity and the society

`APP_USER.language` is `en`, `hi`, or `mr`. The column is a code, so another language is a value, not a migration. `data_notice_at` is the first-open privacy notice. `CONSENT` is a different table and stays empty until

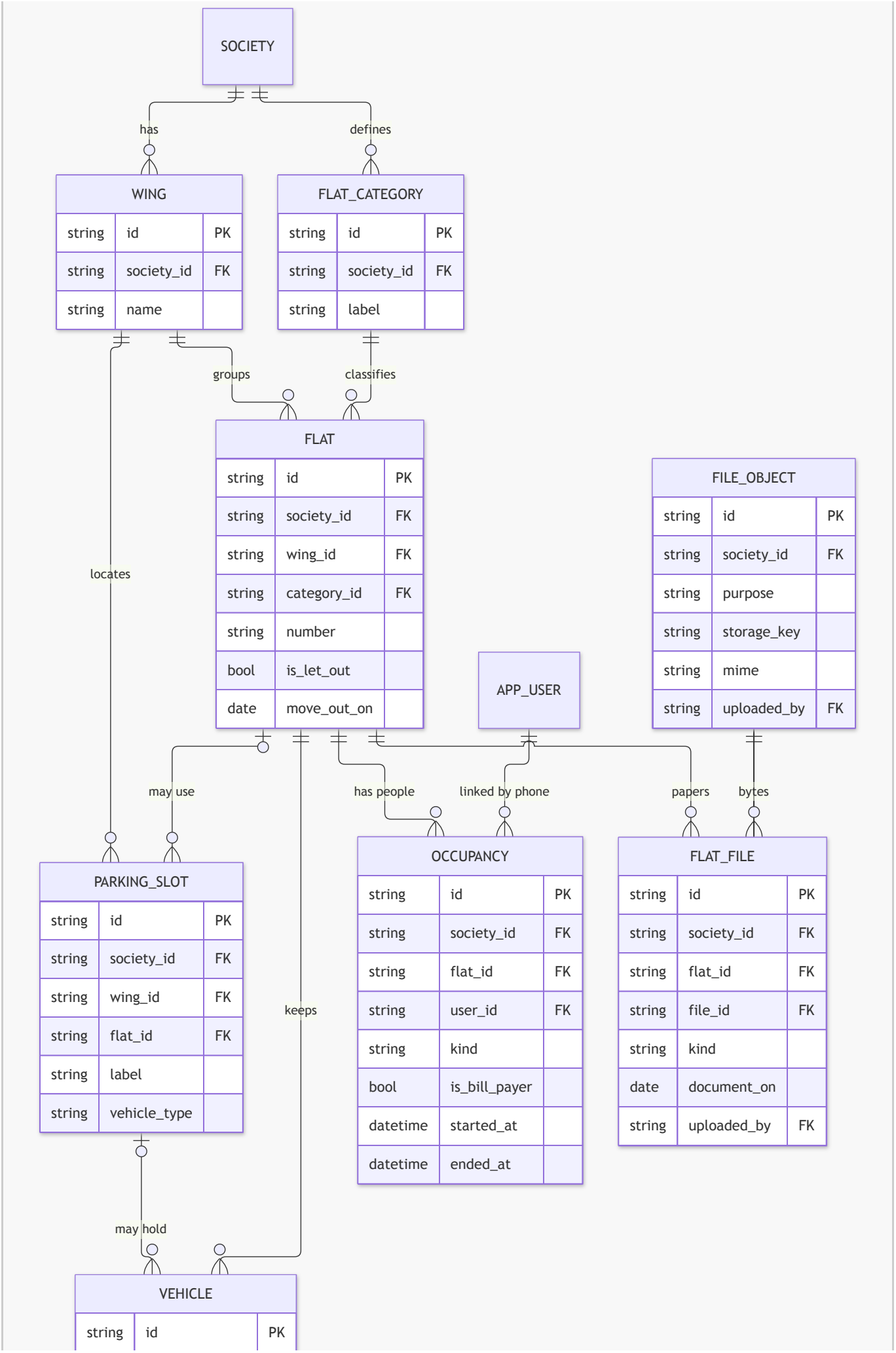
WhatsApp or face check.



- `SOCIETY.status`: `draft`, `active`, `suspended`. New societies start as `draft`.
- `payment_window_days` is 5 to 10. Default 10. That window is the only grace.
- `late_fee_type`: `simple` or `growing`. Default `simple`. `daily_rate_ppm` of 500 means 0.05% per day. `late_fee_cap_paise` is null until the bye-laws are read.
- `upi_vpa` must contain `@`. The app never stores a UPI PIN or a bank password.
- `ROLE_ASSIGNMENT.role`: `system_admin`, `secretary`, `treasurer`, `facility_manager`. Resident access is an `OCCUPANCY`, not a role row. `society_id` is null only for `system_admin`.

- One open assignment per society per committee role. Replacing a person sets `ended_at` on the old row. Old actions keep the old `user_id`.
- `SESSION.society_id` is null only for a system admin. `SESSION.role` is the only role that token may call.
- `FEATURE_FLAG.flag_key`: `services`, `gate`, `face_check`, `whatsapp_auto`. Created off.
- `OTP_CHALLENGE` is not linked to a society. The raw code is not stored.

3. Flats, people, parking, papers

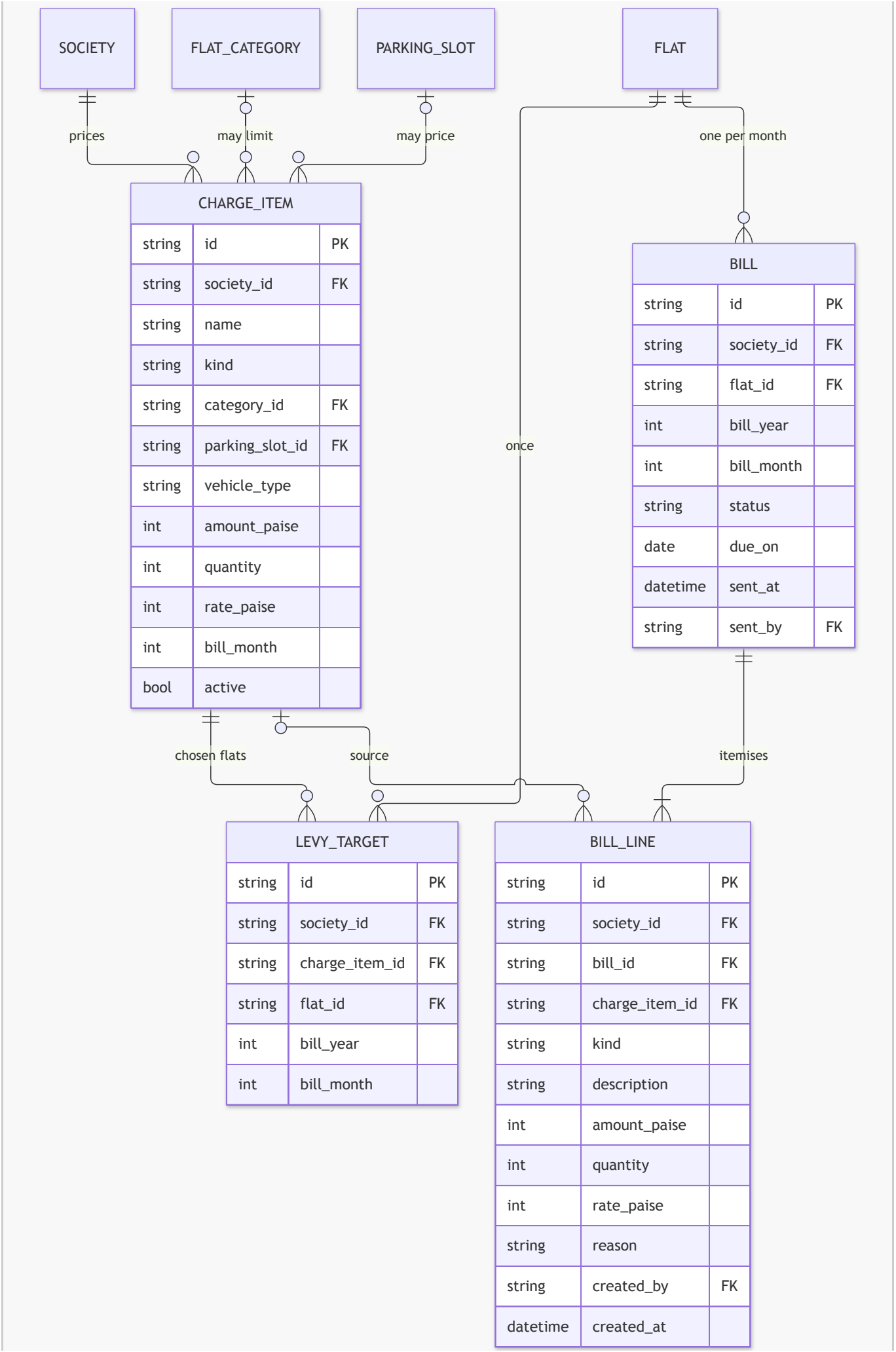

SOCIETY
has
defines
WING
string id PK
string society_id FK
string name
FLAT_CATEGORY
string id PK
string society_id FK
string label
groups
classifies
FLAT
string id PK
string society_id FK
string wing_id FK
string category_id FK
string number
bool is_let_out
date move_out_on
FILE_OBJECT
string id PK
string society_id FK
string purpose
string storage_key
string mime
string uploaded_by FK
APP_USER
locates
may use
has people
linked by phone
papers
bytes
PARKING_SLOT
string id PK
string society_id FK
string wing_id FK
string flat_id FK
string label
string vehicle_type
keeps
OCCUPANCY
string id PK
string society_id FK
string flat_id FK
string user_id FK
string kind
bool is_bill_payer
datetime started_at
datetime ended_at
FLAT_FILE
string id PK
string society_id FK
string flat_id FK
string file_id FK
string kind
date document_on
string uploaded_by FK
may hold
VEHICLE
string id PK

string	society_id	FK
string	flat_id	FK
string	slot_id	FK
string	plate	
string	sticker_number	

- Flat identity is unique on `(society_id, wing_id, number)`.
- `FLAT_CATEGORY.label` starts as 1RK, 1BHK, 2BHK, 3BHK, penthouse, shop. A society can add a label.
- `OCCUPANCY.kind`: `owner`, `tenant`, `extra`. The owner is the bill payer until a secretary switches that flag to the tenant. Partial unique index: one open `is_bill_payer` per flat.
- A phone that is not yet verified still gets an `APP_USER` at import, so reminders have somewhere to go. The first OTP is what sets the language.
- Changing a person's phone updates `APP_USER.mobile`. It does not rewrite old ledger or audit rows. Replacing a person ends the occupancy and starts another. History stays on the old user.
- `PARKING_SLOT.vehicle_type`: `car` or `bike`, or a label the society adds. No barrier, camera, or guard login.
- `FLAT_FILE.kind`: `rental_agreement` or `police_verification`. Readable by the secretary and by occupancies of that flat. Other flats cannot open it. FacilityCore does not perform police verification.
- `FILE_OBJECT` is the only file store. Receipts, screenshots, complaint photos, and the vault point here. Access follows the parent row. Files sit in an Indian region.

4. Charges and bills

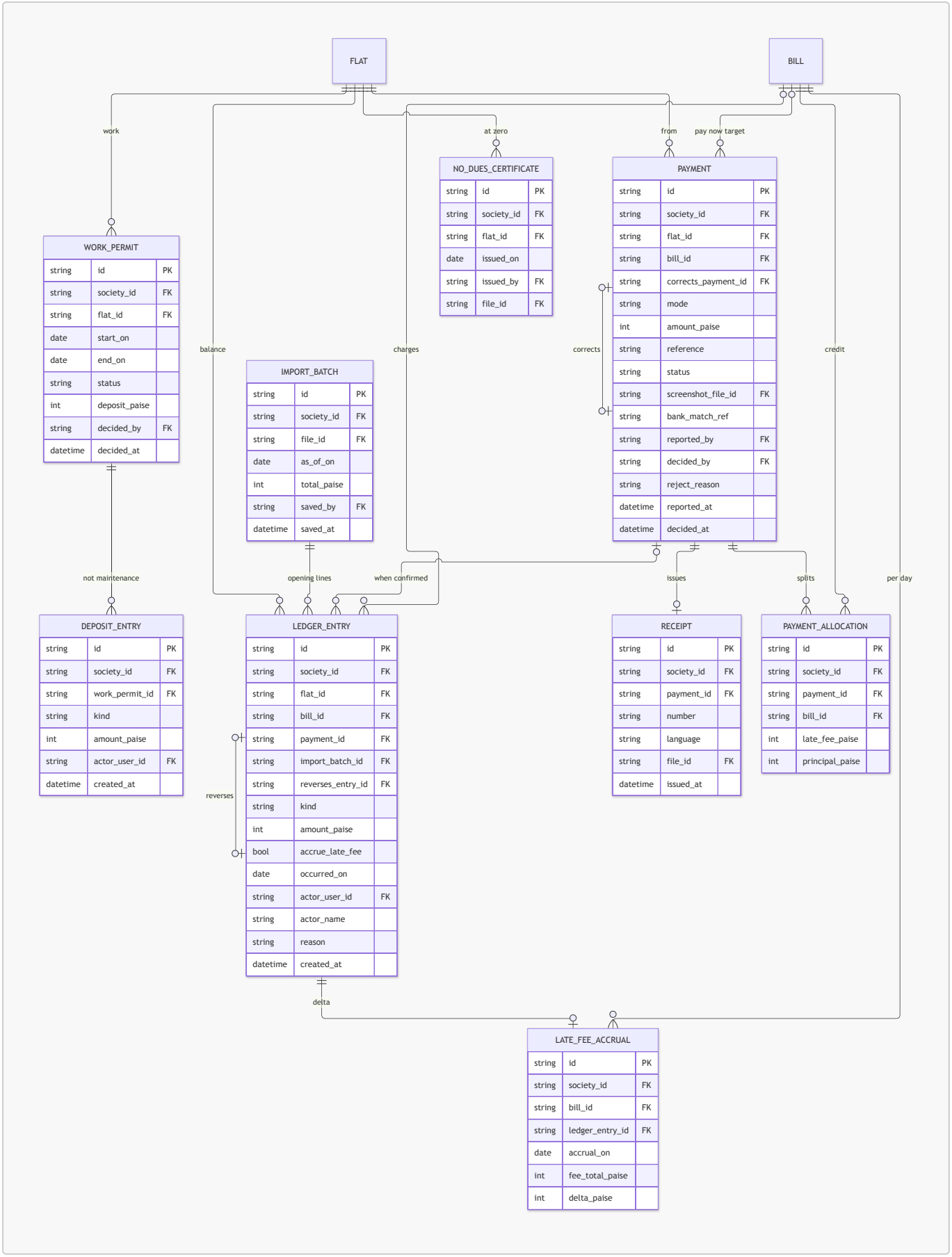
A sent line is a snapshot. Editing a charge item later does not change a bill already sent.


SOCIETY
FLAT_CATEGORY
PARKING_SLOT
FLAT
prices
may limit
may price
one per month
CHARGE_ITEM
string id PK
string society_id FK
string name
string kind
string category_id FK
string parking_slot_id FK
string vehicle_type
int amount_paise
int quantity
int rate_paise
int bill_month
bool active
BILL
string id PK
string society_id FK
string flat_id FK
int bill_year
int bill_month
string status
date due_on
datetime sent_at
string sent_by FK
once
chosen flats
source
itemises
LEVY_TARGET
string id PK
string society_id FK
string charge_item_id FK
string flat_id FK
int bill_year
int bill_month
BILL_LINE
string id PK
string society_id FK
string bill_id FK
string charge_item_id FK
string kind
string description
int amount_paise
int quantity
int rate_paise
string reason
string created_by FK
datetime created_at

- `CHARGE_ITEM.kind`: `monthly`, `annual`, `fund`, `levy`, `parking`, `non_occupancy`.
- `category_id` null means every flat. Set it to charge a 2BHK differently from a shop.
- `bill_month` is used when `kind` is `annual`. A levy uses `LEVY_TARGET` instead, one row per chosen flat for that month.
- `quantity` and `rate_paise` stay null until a later water-meter charge fills them. The bill line copies them.
- Parking: if `parking_slot_id` is set, the line is added only to that slot's flat. If `kind` is `parking`, the slot is null, and `vehicle_type` is set, the run adds the amount once per linked slot of that type on the flat.
- Non-occupancy ships inactive. The run adds it only when the item is active and `FLAT.is_let_out` is true. No legal percentage is built in.
- `BILL` is unique on `(society_id, flat_id, bill_year, bill_month)`. A second run does nothing for a flat that already has that month.
- `BILL.status`: `draft`, `sent`, `partially_paid`, `awaiting_confirmation`, `paid`. Residents cannot read a draft.
- Status is updated in this order: a `reported` payment makes it `awaiting_confirmation`; balance 0 makes it `paid`; some credit and a remaining balance makes it `partially_paid`; otherwise `sent`.
- `due_on` is the bill date plus `payment_window_days`. Late fees start the next day.
- An adjustment inserts a `BILL_LINE` and a `LEDGER_ENTRY`. The original lines stay. `reason` and `created_by` are required on an adjustment.

5. Ledger, payments, deposits

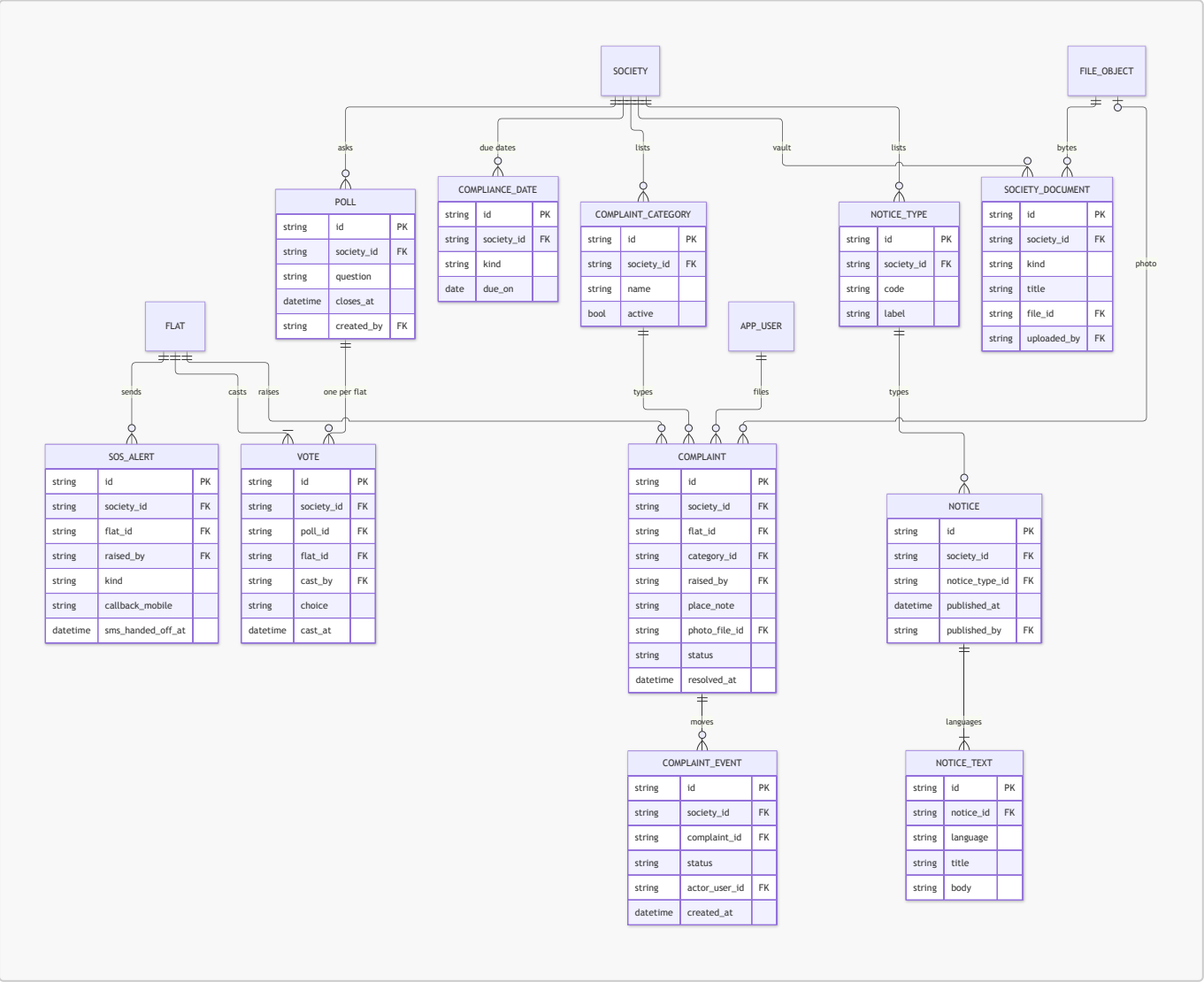
This is the money model. The flat balance is the sum of its ledger. Interior deposits are not in that sum.



- `LEDGER_ENTRY.kind`: `opening`, `charge`, `late_fee`, `payment`, `adjustment`, `waiver`, `bounce`, `bounce_charge`, `refund`.
- Opening line: positive if the flat owes, negative if it is in advance. `accrue_late_fee` is false unless the secretary opts in for that flat. Charge lines always accrue.
- `PAYMENT.mode`: `upi`, `cash`, `cheque`. `PAYMENT.status`: `reported`, `confirmed`, `rejected`.

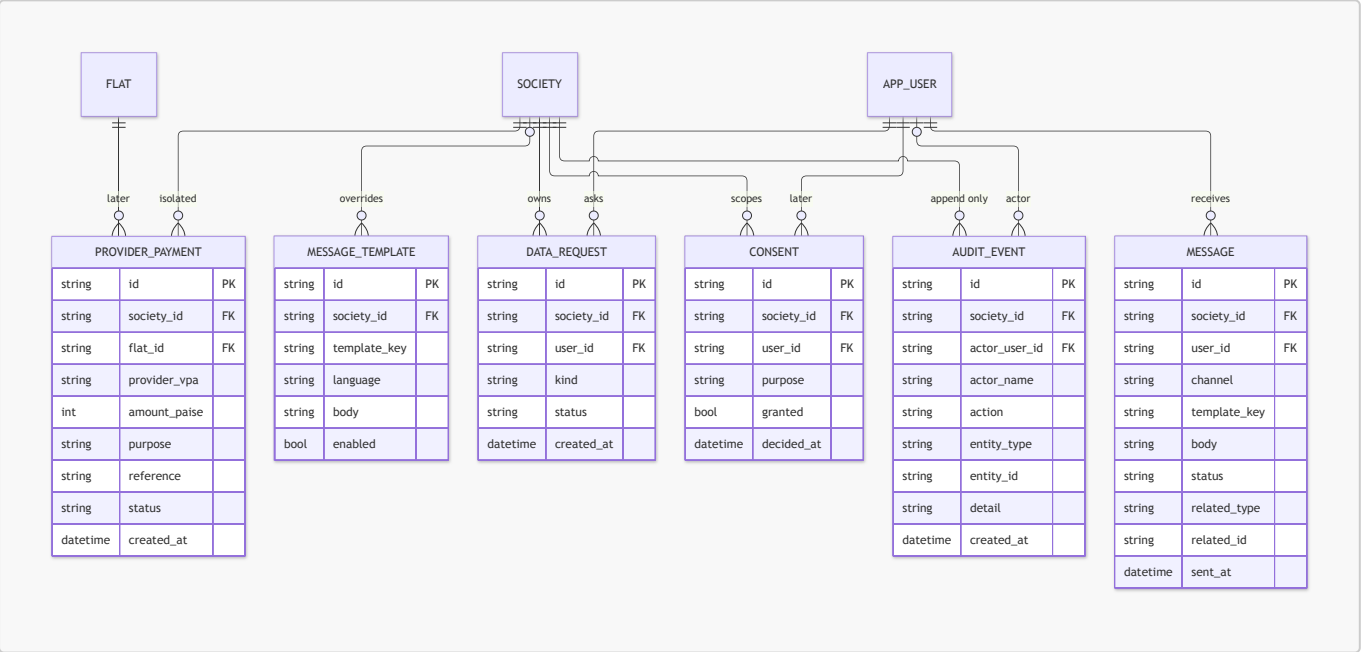
- Unique UPI reference per society among `reported` and `confirmed`. Cheque numbers are stored and are not under that unique rule. Cash may have a null reference.
- Confirm writes the payment ledger row (negative), then `PAYMENT_ALLOCATION` rows: late fees first, then oldest unpaid bill. `bank_match_ref` stays null. Matching is manual in this phase.
- A bounced cheque writes `bounce` for the same amount (positive) and may write `bounce_charge`. Late fees use the original `due_on`.
- A payment on the wrong flat is not moved. The ledger is reversed with a reason, and a new payment is written on the right flat through `corrects_payment_id`.
- `LATE_FEE_ACCRUAL` is unique on `(bill_id, accrual_on)`. The job stores that day's fee total and posts only `delta_paise`, so a rerun does not double-charge.
- Simple fee total = unpaid principal × daily rate × late days. Growing amount due = principal × (1 + daily rate) ^ late days. The growing check is Rs 3,000 at 500 ppm: Rs 3,001.50 on the first late day, Rs 3,015.03 after 10 late days. A cap, when set, stops the fee. Waivers are `waiver` rows. Only a secretary can write one, and `reason` is required.
- `WORK_PERMIT.status`: `submitted`, `approved`, `refused`. `DEPOSIT_ENTRY.kind`: `collect`, `refund`. No foreign key from either table to `LEDGER_ENTRY`.
- A no-dues certificate can be inserted only when the flat's ledger sum is 0. Dues stay on the flat after `move_out_on`.

6. Helpdesk, community, documents



- Complaint categories are seeded per society: plumbing, electrical, lift, drainage choke, seepage, water supply, DG and power backup, parking, noise, common-area cleaning. The secretary can add one. No release required.
- `COMPLAINT.status`: `pending`, `in_progress`, `resolved`. The guard is not a recipient. Within 7 days of `resolved_at` the resident can reopen, which appends a `pending` event.
- Notice types are seeded: water, tanker, lift, meeting, general. A notice does not create a bill and does not collect a vote. `NOTICE_TEXT` holds en, hi, and mr. The home screen uses the resident's language.
- `VOTE` is unique on `(poll_id, flat_id)`. The open bill payer casts it. Choice is `yes` or `no`.
- `SOS_ALERT.kind`: `medical`, `fire`, `security`. SMS goes to the open secretary, the open facility manager, and `duty_guard_mobile`, with flat number and callback phone. `sms_handed_off_at` minus `created_at` is tested against 10 seconds.
- `SOCIETY_DOCUMENT.kind`: `bye_laws`, `occupancy_certificate`, `fire_noc`, `lift_licence`, `insurance`, `minutes`. Any resident of that society can read them.
- `COMPLIANCE_DATE.kind`: `lift_amc`, `fire_noc`, `building_insurance`, `agm`. The dashboard lists rows due within 30 days. This does not book a job.

7. Messages, privacy, and the seams



- `MESSAGE_TEMPLATE.template_key`: `otp`, `bill_sent`, `reminder_before`, `reminder_due`, `late_started`, `weekly_balance`, `complaint_status`, `payment_rejected`, `sos`. One row per language. `society_id` null is the platform default. A society row overrides it. The secretary can set `enabled` false. OTP uses the same SMS path and cannot be switched off.
- `MESSAGE.channel`: `sms` or `in_app`. Share-on-WhatsApp is not a row: the phone opens WhatsApp and the resident sends it. Automatic WhatsApp is the `whatsapp_auto` flag, which is off, and a future `CONSENT.purpose` of `whatsapp_reminders`.
- `DATA_REQUEST.kind`: `see_my_data` or `society_export`. It is stored for the grievance contact. Profile shows `SOCIETY.grievance_name`.
- `CONSENT.purpose` later: `whatsapp_reminders`, `face_check`. No rows in this phase.
- `PROVIDER_PAYMENT.provider_vpa` is not `SOCIETY.upi_vpa`. Insert is allowed for a later priced service. Any API that would allocate it to a bill or a ledger row is refused.

- AUDIT_EVENT has no update and no delete. Reject, role grant, suspend, handover, waiver, and bill send are all rows here. actor_name is copied at write time so a later rename does not rewrite history.
- An opened bill may be cached on the phone. Pay, complaints, and SOS need a connection. Backup is an operations job, every day, kept 30 days. It is not a table.

Where later work attaches

Later work	Uses rows that exist now	Does not get a table yet
Priced home services	PROVIDER_PAYMENT, flag services	vendor, catalogue, slot booking
Society services paid back through dues	a levy charge item	a second ledger
Automatic WhatsApp	CONSENT, MESSAGE, flag whatsapp_auto	a WhatsApp API call in this phase
Gate, after a poll	VEHICLE, PARKING_SLOT, FLAT_FILE, flag gate	visitor, guest code, guard login
Face check, after a privacy review	CONSENT, flag face_check	face template
Metered water	quantity and rate_paise on the charge and the line	a meter table

Held out, with no extension point: local vendors and the 2 km directory, a payment gateway, commission, NACH, GST returns, payroll, redevelopment, rent receipts for HRA, Aadhaar face match, boom barriers.

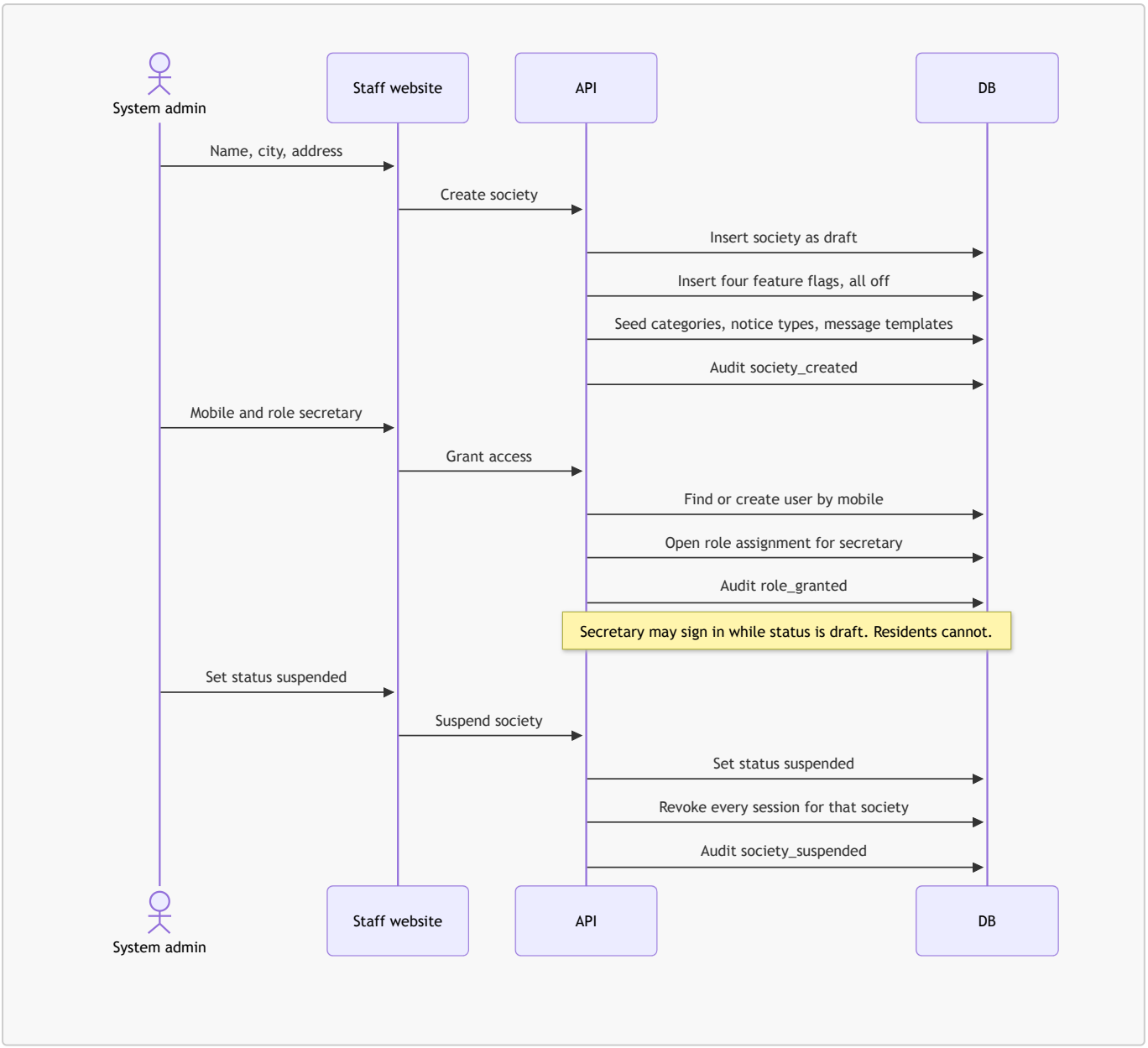
8. Coverage

Feature	Stored as
F36 Onboard a society	SOCIETY, FEATURE_FLAG, AUDIT_EVENT
F37 Committee access	ROLE_ASSIGNMENT with ended_at
F01 OTP, roles, language	OTP_CHALLENGE, SESSION, APP_USER.language, APP_USER.data_notice_at
F02 Rules and UPI ID	columns on SOCIETY
F03 Flats, people, Excel	FLAT, OCCUPANCY, APP_USER, IMPORT_BATCH
F04 Opening balances	LEDGER_ENTRY.kind = opening, accrue_late_fee
F05 Wings and categories	WING, FLAT_CATEGORY
F06 Parking and vehicles	PARKING_SLOT, VEHICLE
F07 Handover	close and open ROLE_ASSIGNMENT; audit keeps the old actor
F08 Charge catalogue	CHARGE_ITEM, snapshot on BILL_LINE

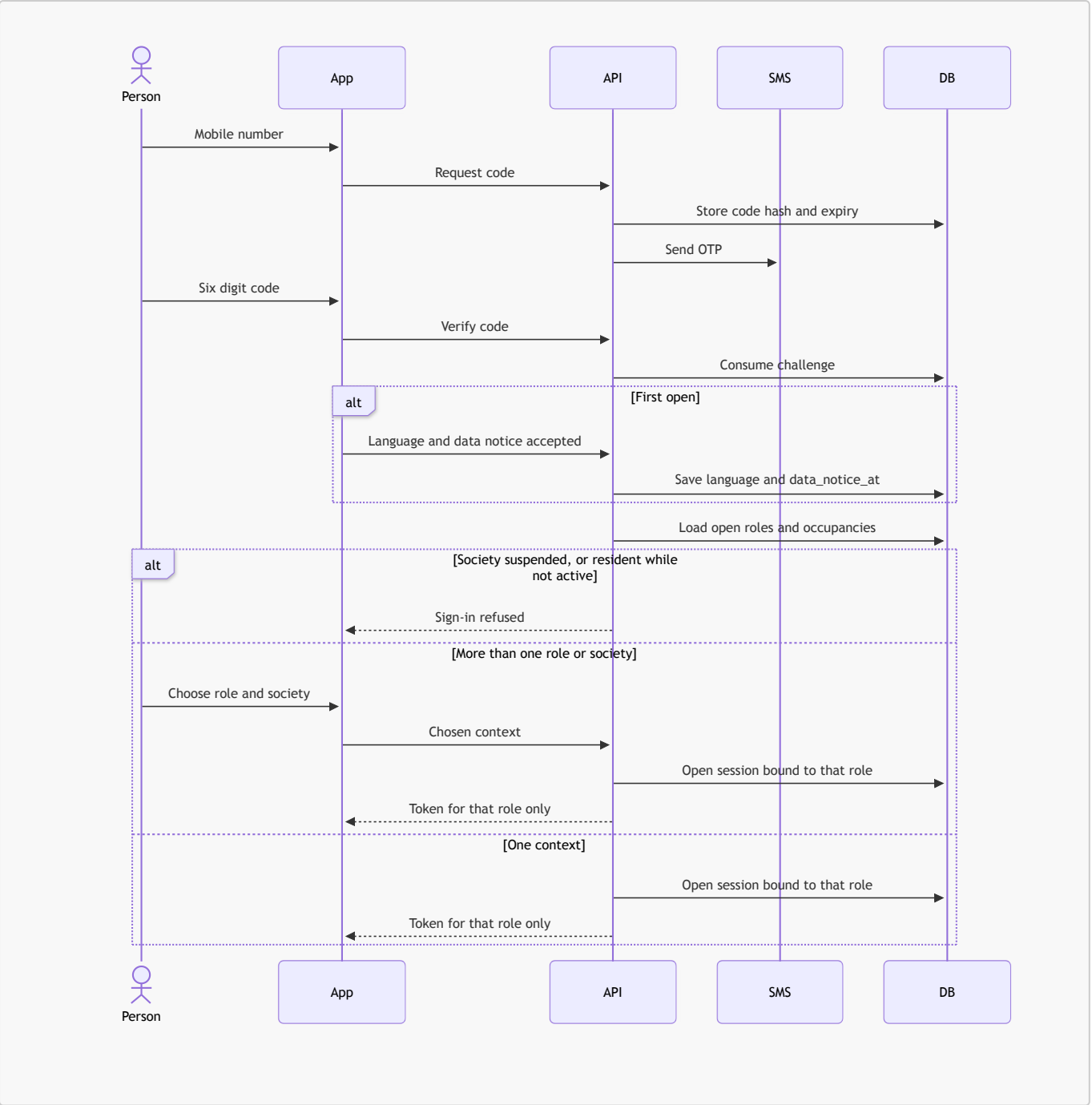
Feature	Stored as
F09 Funds, levies, non-occupancy, parking	`CHARGE_ITEM.kind`, `LEVY_TARGET`, `FLAT.is_let_out`
F10 Draft, review, send, correct	`BILL`, `BILL_LINE` adjustment
F11 No double charge	unique `(society_id, flat_id, bill_year, bill_month)`
F12 UPI intent and QR	no table; `SOCIETY.upi_vpa` and the amount due
F13 UTR, confirm, reject	`PAYMENT`, ledger only on confirm
F14 Block a reused UTR	unique live reference per society
F15 Cash, cheque, repairs	`PAYMENT.mode`, `bounce`, `refund`, `corrects_payment_id`
F16 Statement and receipt	ledger list, `RECEIPT`
F17 Move-out and no-dues	`FLAT.move_out_on`, `NO_DUES_CERTIFICATE`
F18 Tenancy papers	`FLAT_FILE`
F19 Interior deposit	`WORK_PERMIT`, `DEPOSIT_ENTRY`, no ledger FK
F20 Late fee and waiver	`LATE_FEE_ACCRUAL`, `LEDGER_ENTRY.kind = waiver`
F21 SMS and in-app reminders	`MESSAGE_TEMPLATE`, `MESSAGE`
F22 Share on WhatsApp	no row; the resident sends it
F23 Defaulter list	a query on ledger and due date; resident token refused
F24 Helpdesk and SLA	`COMPLAINT`, `COMPLAINT_EVENT`
F25 Complaint categories	`COMPLAINT_CATEGORY`
F26 Local vendors	no table
F27 Polls	`POLL`, `VOTE` unique per flat
F28 SOS	`SOS_ALERT`, `SOCIETY.duty_guard_mobile`
F29 Notices	`NOTICE`, `NOTICE_TEXT`, `NOTICE_TYPE`
F30 Vault and due dates	`SOCIETY_DOCUMENT`, `COMPLIANCE_DATE`
F31 Dashboard, Excel, history	queries plus `AUDIT_EVENT` and `LEDGER_ENTRY`
F32 April–March export	same rows, filtered; fund kinds totalled apart
F33 Backup, speed, roles	backup is operational; role is `SESSION.role`
F34 Privacy	`DATA_REQUEST`, `grievance_name`, Indian file region
F35 Flags and the service seam	`FEATURE_FLAG`, `PROVIDER_PAYMENT`, `CONSENT`, shared `FILE_OBJECT`

9. Sequences

Onboard a society and name who runs it

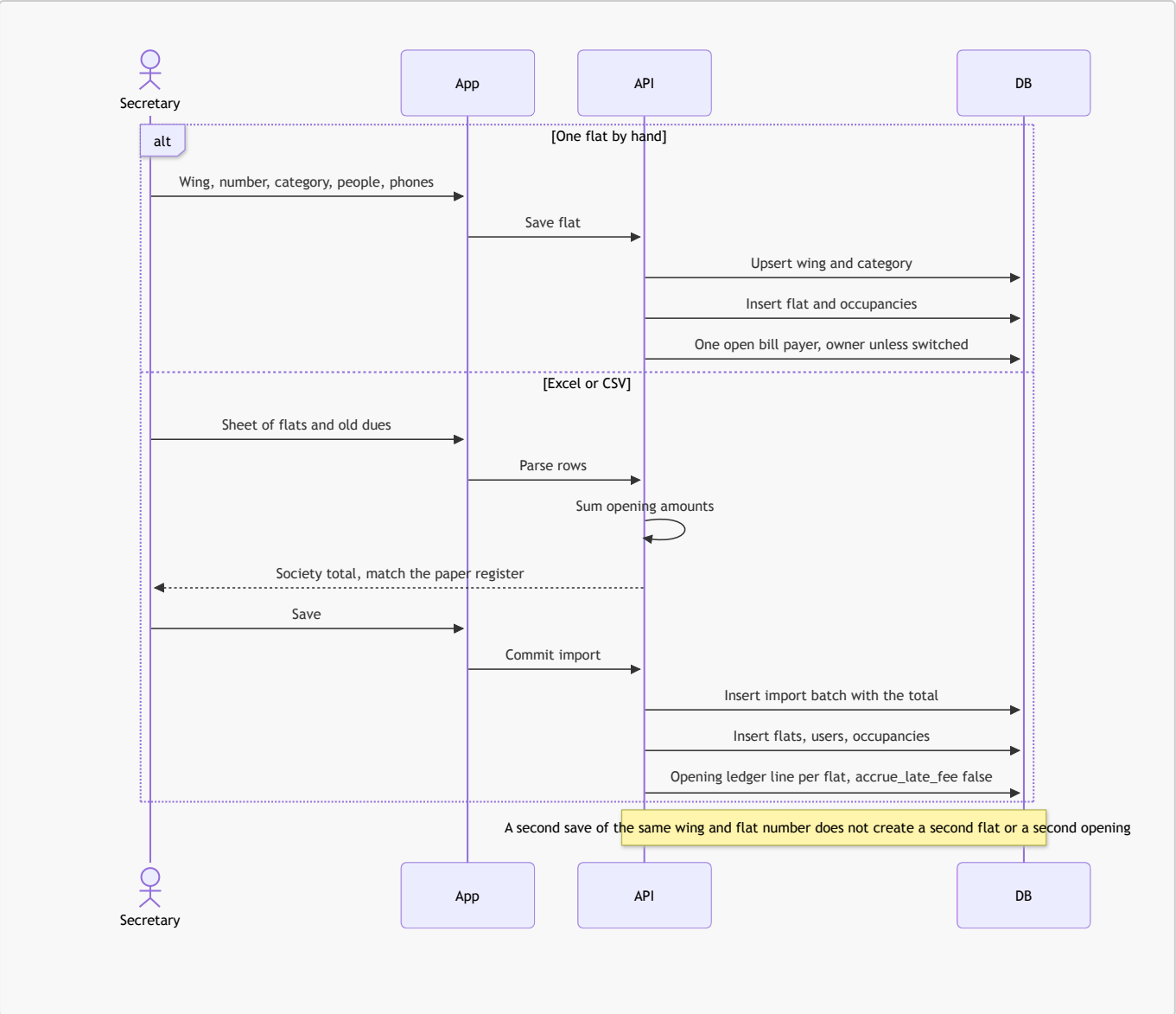


Sign in, language, and one role per token



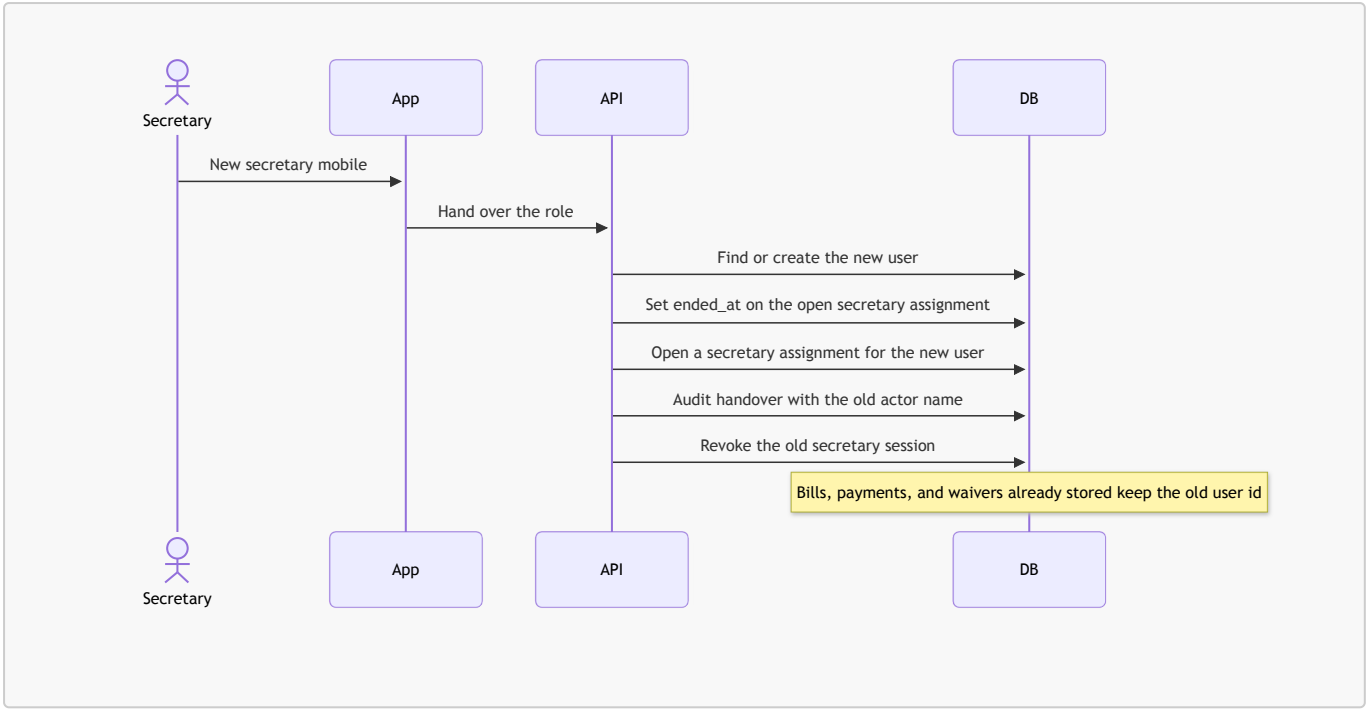
Profile can change `language` later. The same phone in a new role asks for a new token. Logout sets `revoked_at`.

Flats, people, and opening balances

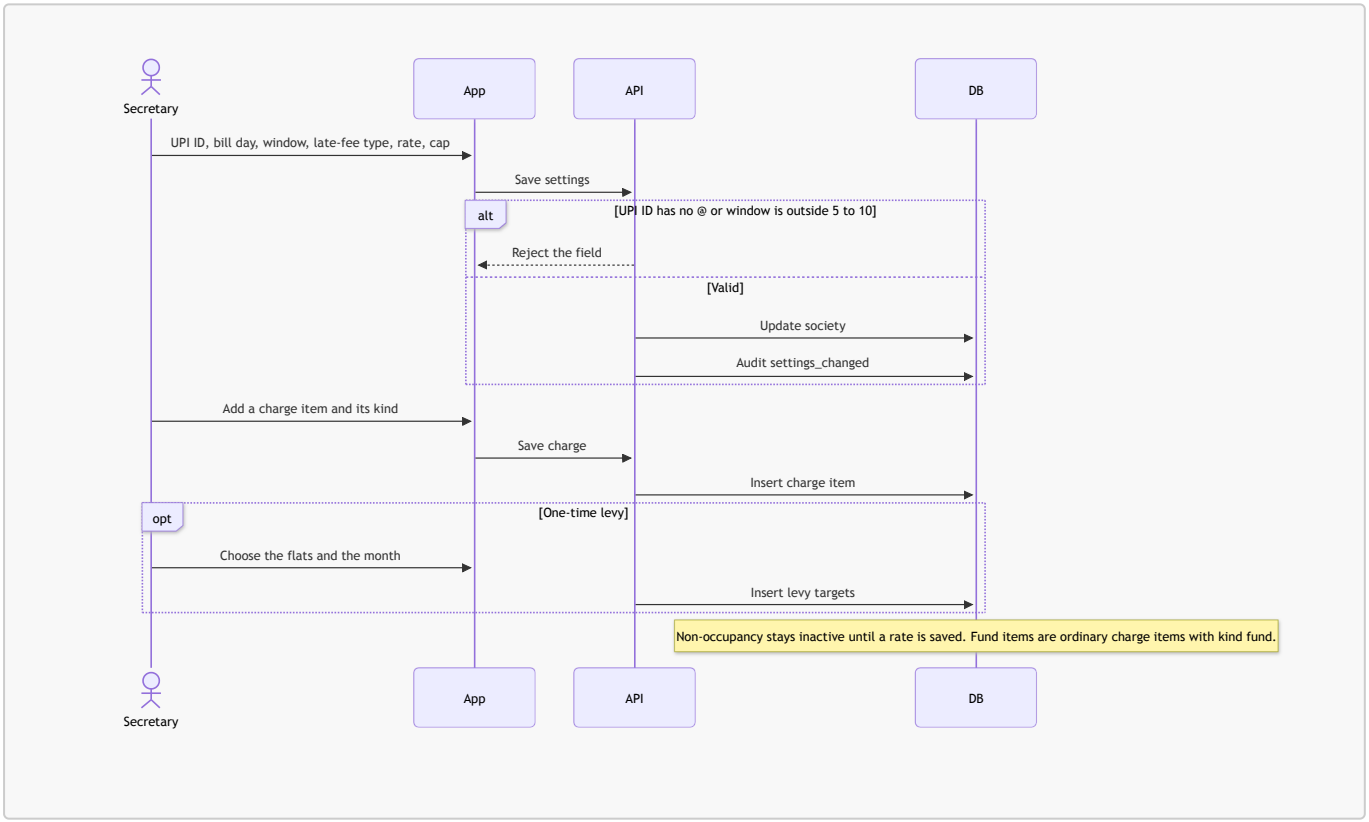


Opting a flat in for late fees on the old due sets `accrue_late_fee` on that opening line only.

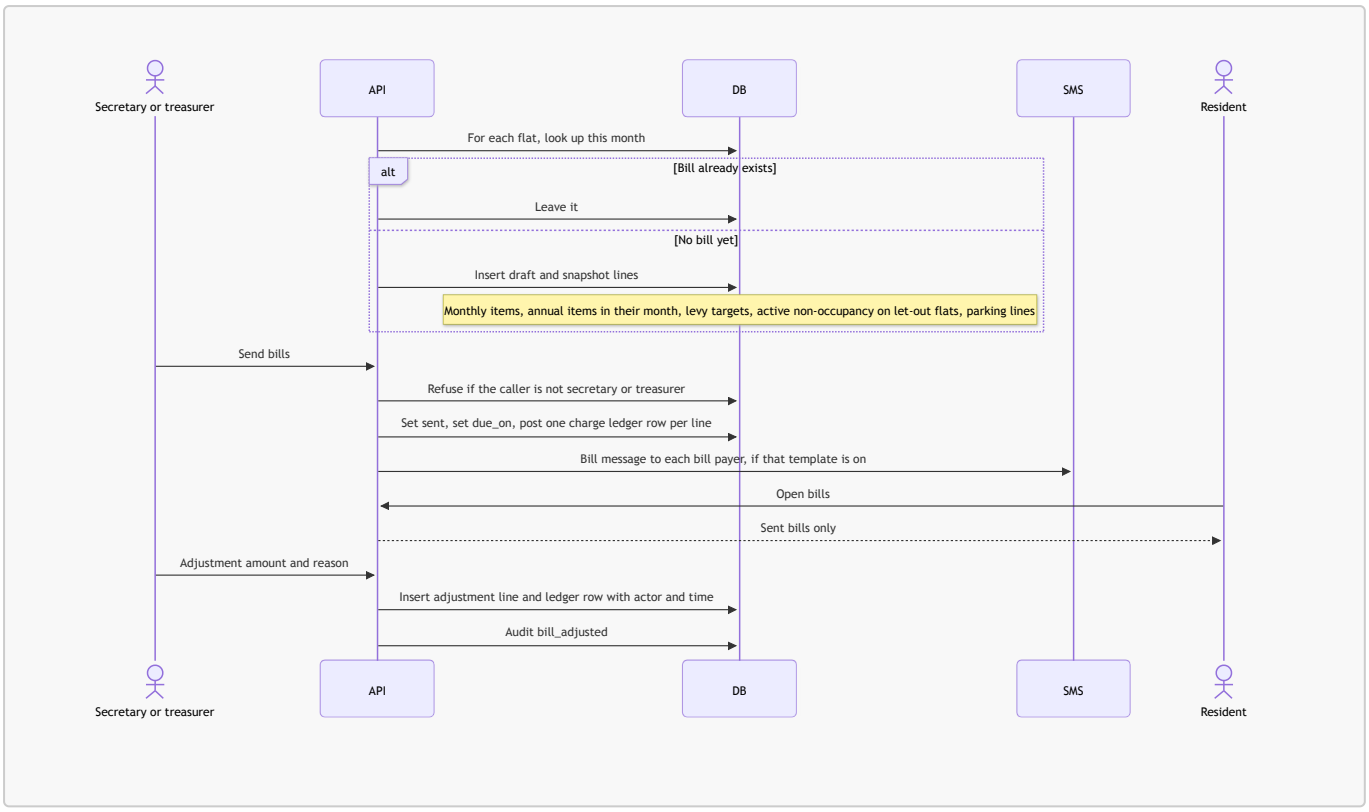
Handover



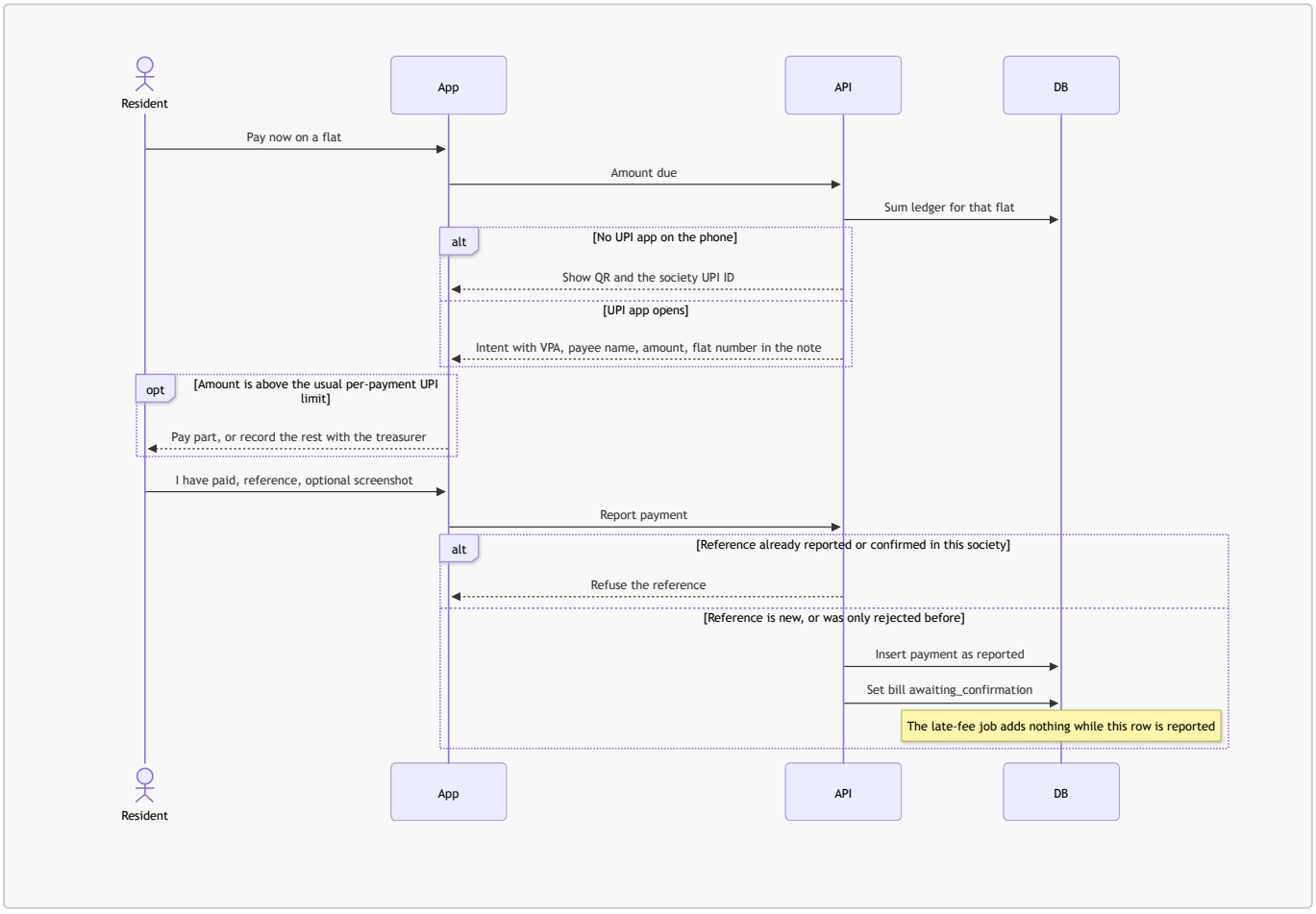
Settings and charges



Draft, send, and correct a bill

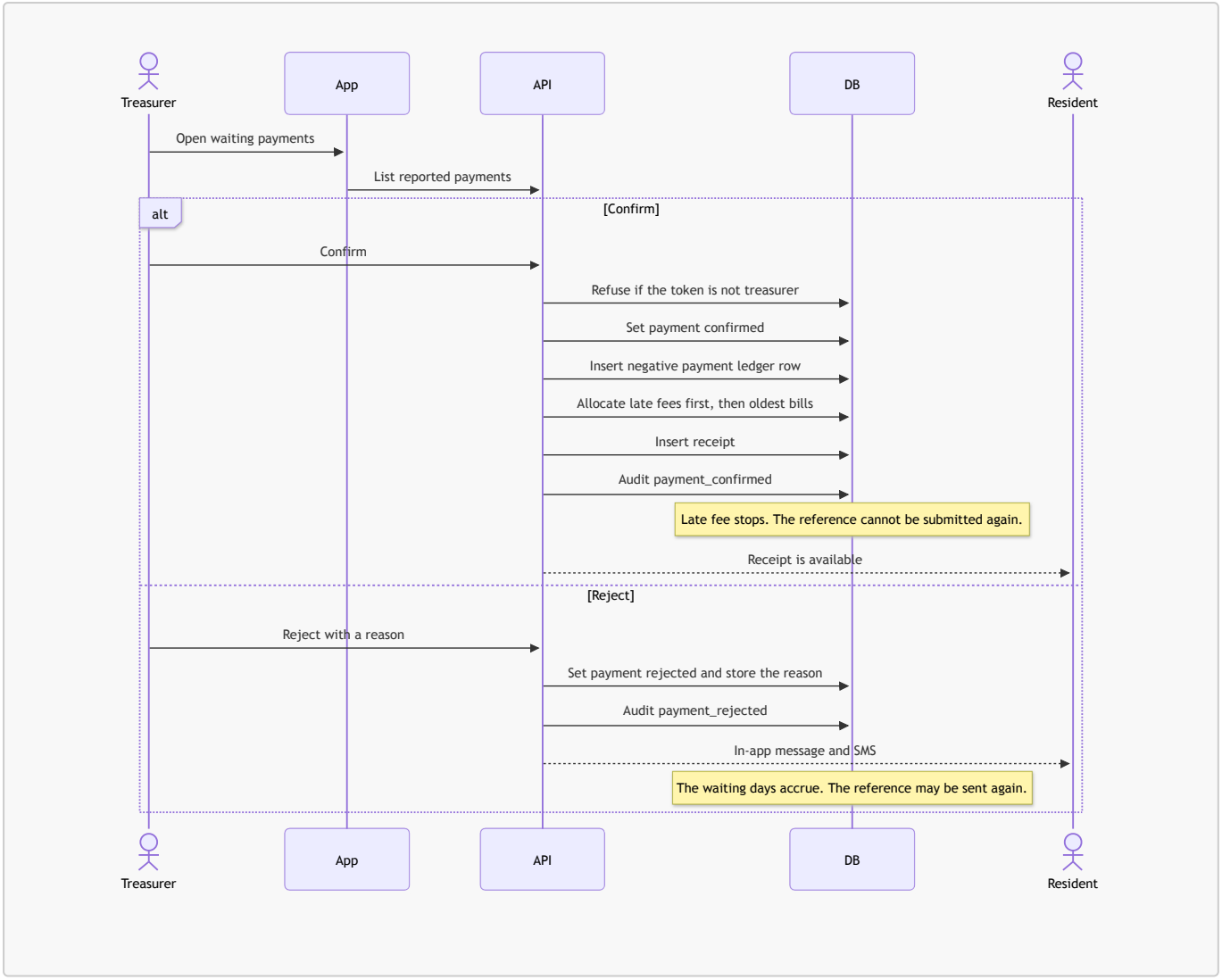


Pay by UPI and report the reference

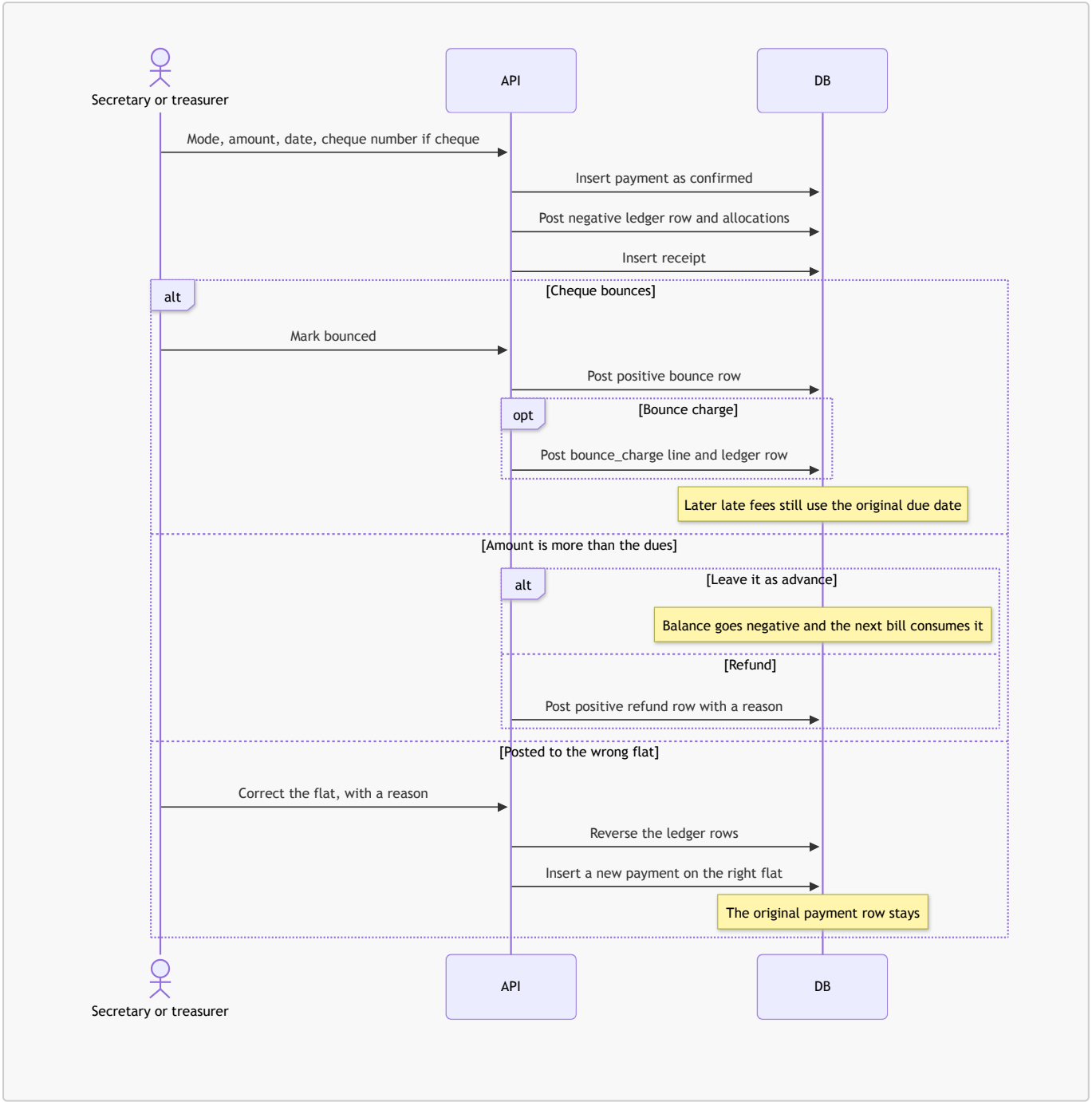


Home is derived. Balance above zero and no reported payment: due. A reported payment: waiting. Balance at or below zero: clear.

Confirm or reject

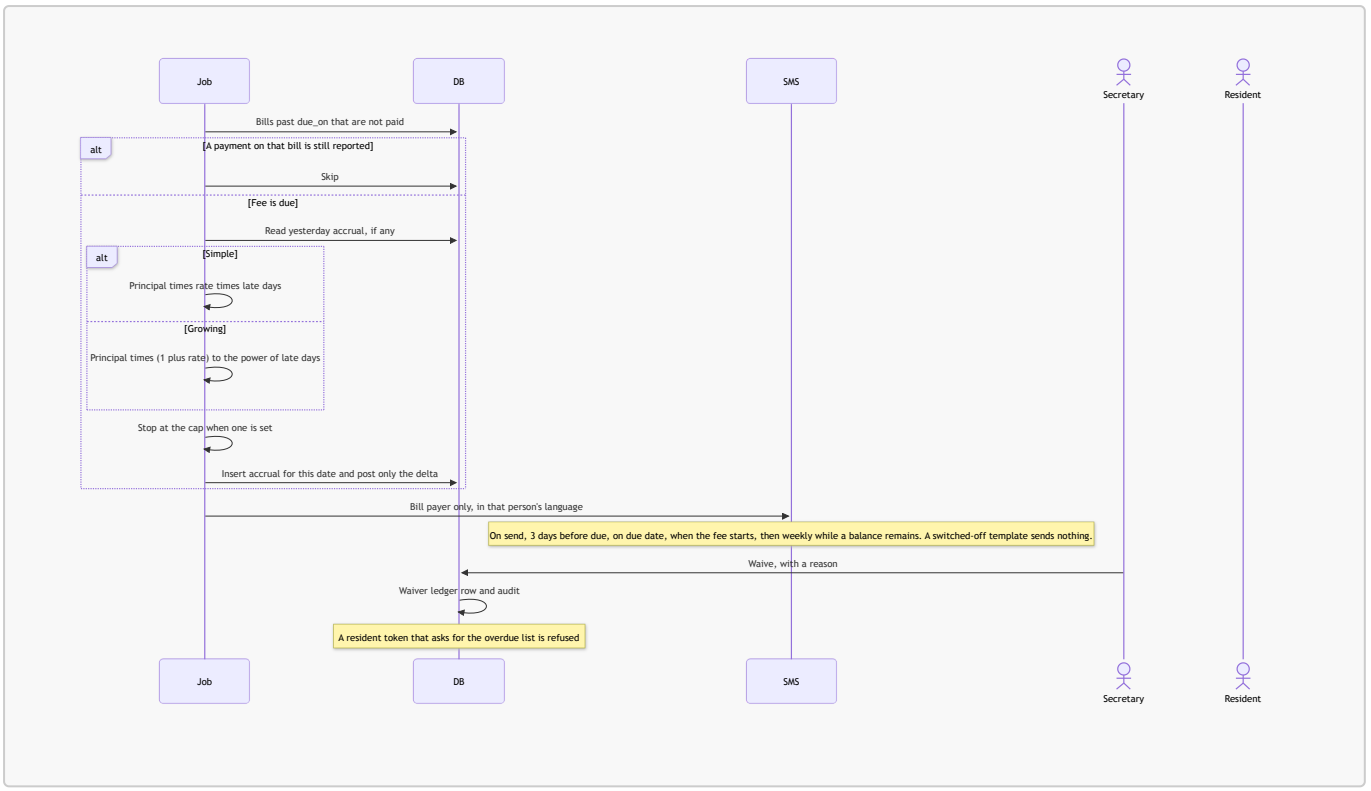


Cash, cheque, bounce, advance, and a payment on the wrong flat



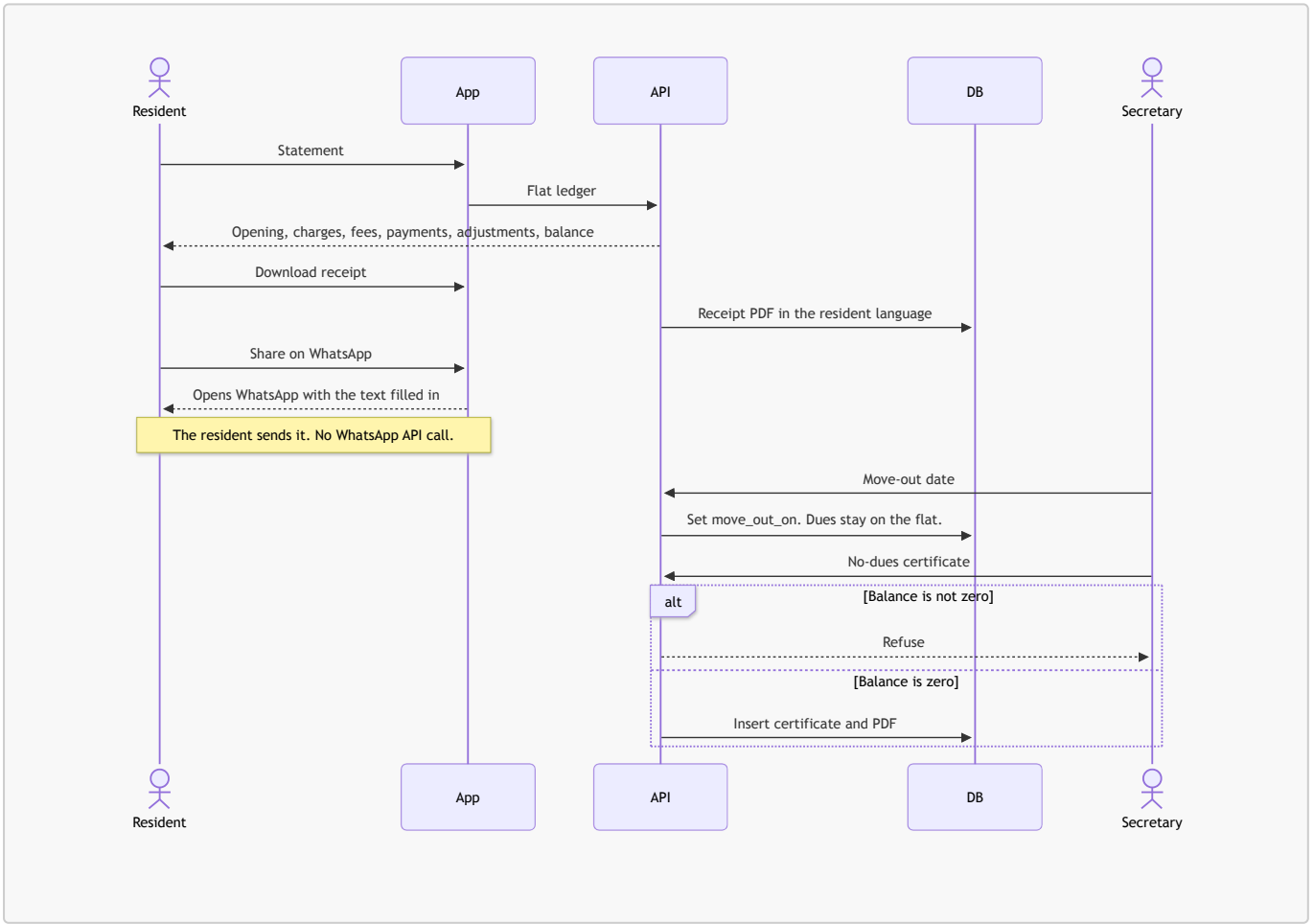
A short payment leaves a positive balance and the bill stays `partially_paid`.

Late fees, reminders, and the defaulter list

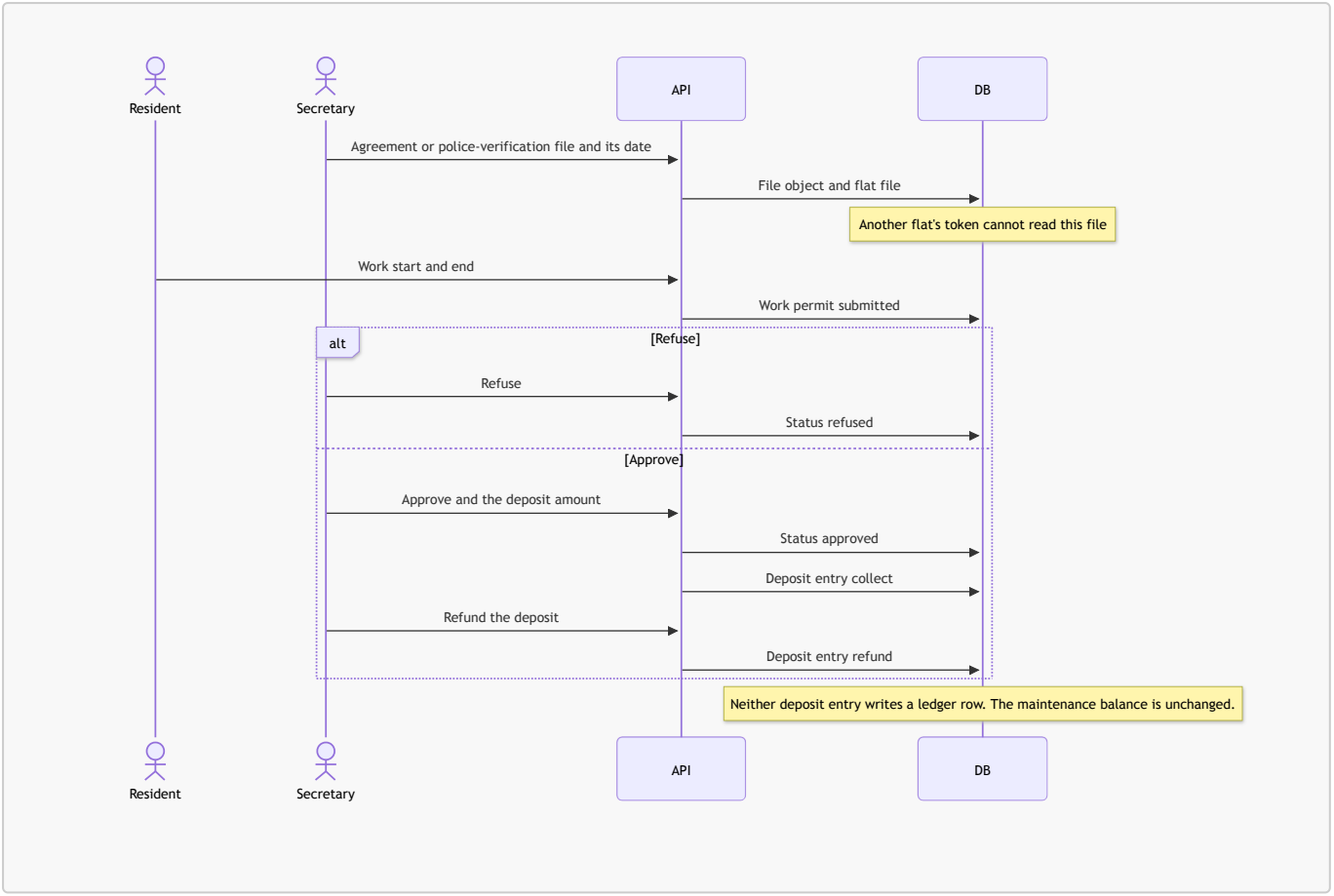


The opening balance is included in the principal only when that ledger row has `accrue_late_fee`.

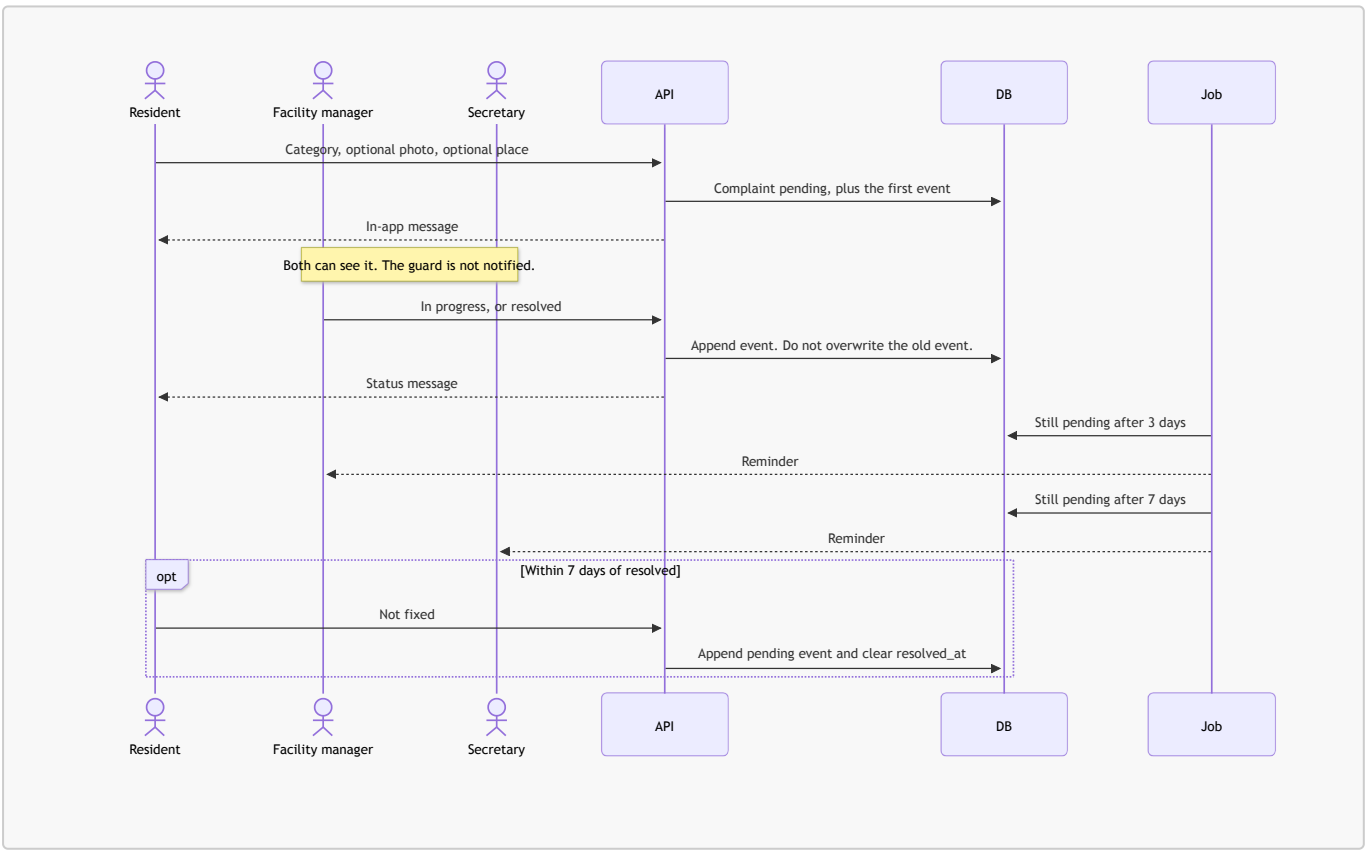
Statement, receipt, share, move-out



Tenancy papers and the interior deposit

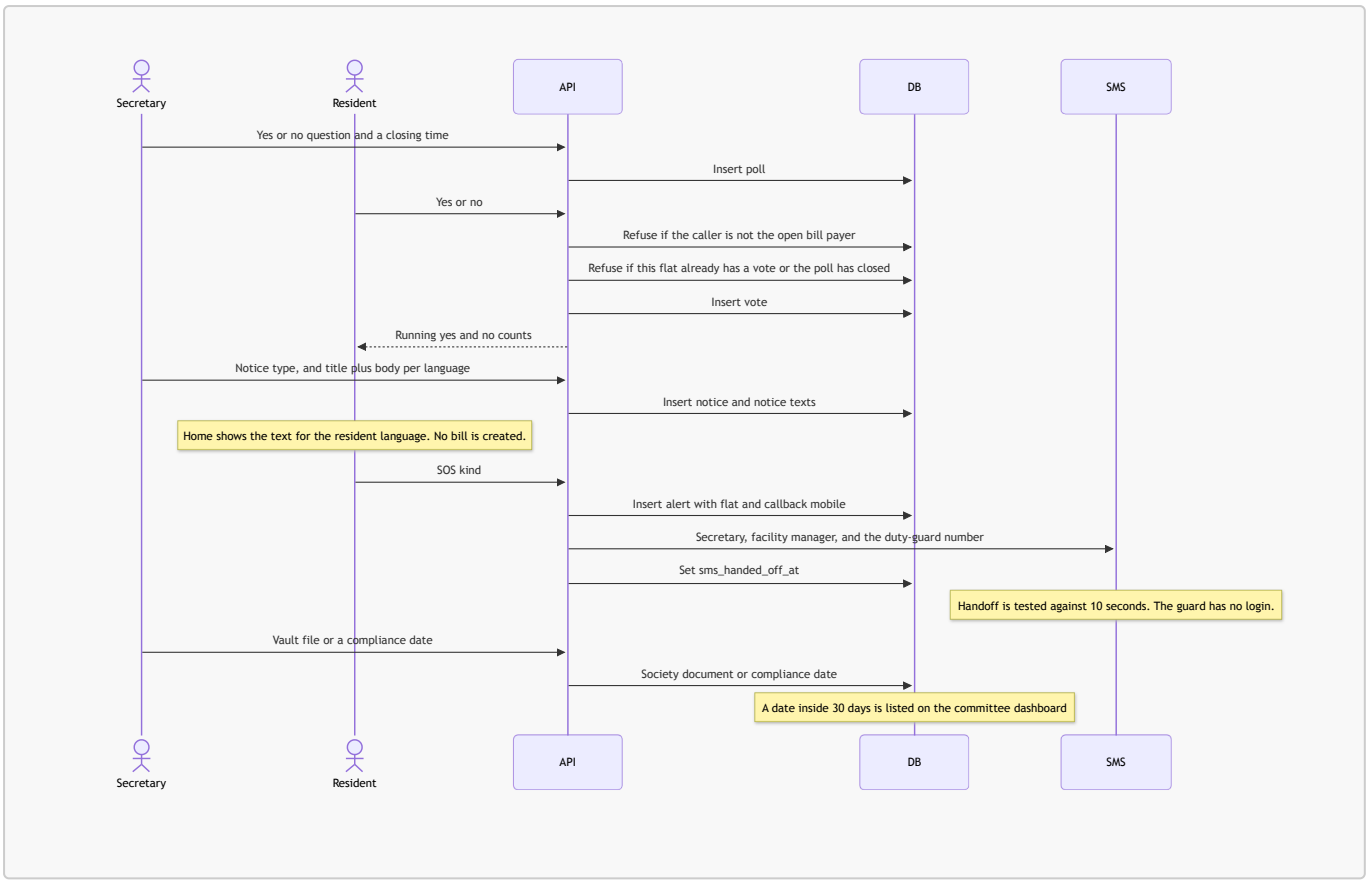


Complaints



The secretary can add a complaint category. That is an insert into `COMPLAINT_CATEGORY`.

Polls, notices, SOS, and the vault



Reports, history, privacy, and service money

